

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: *Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3,BL4,BL5)*

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 25

Total Marks: 50

Annexure A

Design and Develop Model

DURATION: 4 HRS (2 Sessions)

Model-Based RL Approach

1. Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.
(5 Marks)
2. Draw a concept map showing the relationship between the Environment, Model, Planning Module, Policy, and Reward in Model-Based Reinforcement Learning.
(5 Marks)
3. Design a flowchart explaining how an RL agent learns an environment model before selecting an optimal action.
(5 Marks)
4. Draw the architecture of a Model-Based RL system showing model learning, planning, and execution components.
(5 Marks)

5. Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.

(5 Marks)

Analytic Gradient Computation

6. Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.

(5 Marks)

7. Design a concept map showing how gradients are computed and propagated during policy learning.

(5 Marks)

8. Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.

(5 Marks)

9. Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.

(5 Marks)

10. Draw a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.

(5 Marks)

Sampling-Based Planning

11. Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.

(5 Marks)

12. Prepare a concept map showing how sampled trajectories are generated and evaluated for planning.

(5 Marks)

13. Draw a block diagram showing the interaction between the environment model, planner, sampled actions, and optimal policy.

(5 Marks)

14. Design a flowchart explaining Monte Carlo sampling for planning in Reinforcement Learning.

(5 Marks)

15. Draw a concept map comparing exhaustive search planning and sampling-based planning techniques.

(5 Marks)

Model-Based Data Generation

16. Draw a flowchart illustrating synthetic data generation using a learned environment model.

(5 Marks)

17. Prepare a concept map showing the relationship among environment model, generated experiences, replay buffer, and policy learning.

(5 Marks)

18. Draw a block diagram explaining model-generated experience replay in Reinforcement Learning.

(5 Marks)

19. Design a flowchart showing how simulated experiences improve sample efficiency during training.

(5 Marks)

20. Draw a concept map illustrating the advantages and limitations of model-generated training data.

(5 Marks)

Value-Equivalence Prediction and Policy Optimization

21. Draw a flowchart explaining value-equivalence prediction for estimating future rewards.

(5 Marks)

22. Prepare a concept map showing the relationship among value functions, reward prediction, policy evaluation, and policy improvement.

(5 Marks)

23. Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.

(5 Marks)

24. Design a flowchart explaining iterative policy optimization using learned environment dynamics.

(5 Marks)

25. Draw a comprehensive concept map integrating Model-Based RL, Analytic Gradient Computation, Sampling-Based Planning, Model-Based Data Generation, Value-Equivalence Prediction, and Model-Based Policy Optimization into a unified Reinforcement Learning framework.

(5 Marks)

Code of Conduct:

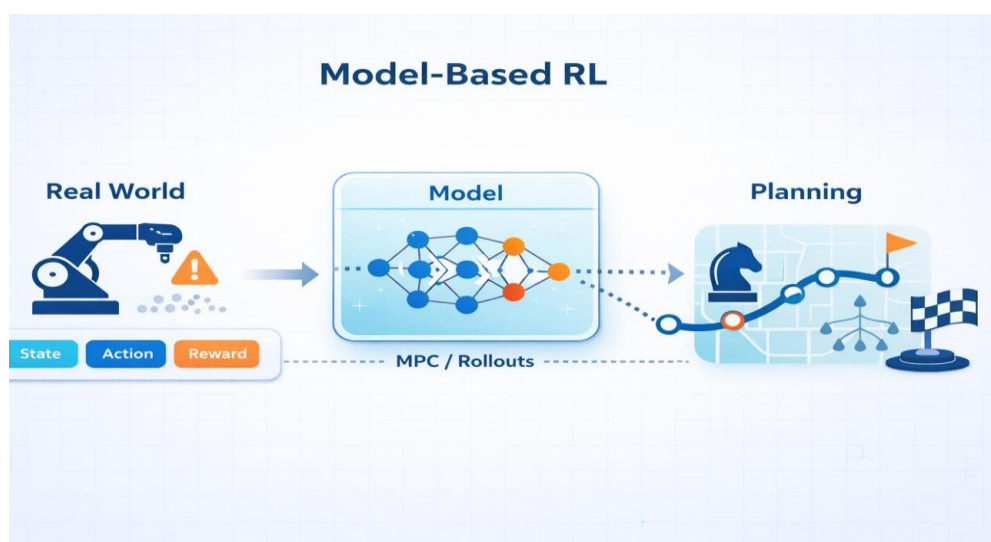
*I **Dasari Haritha(192425254)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Dasari Haritha

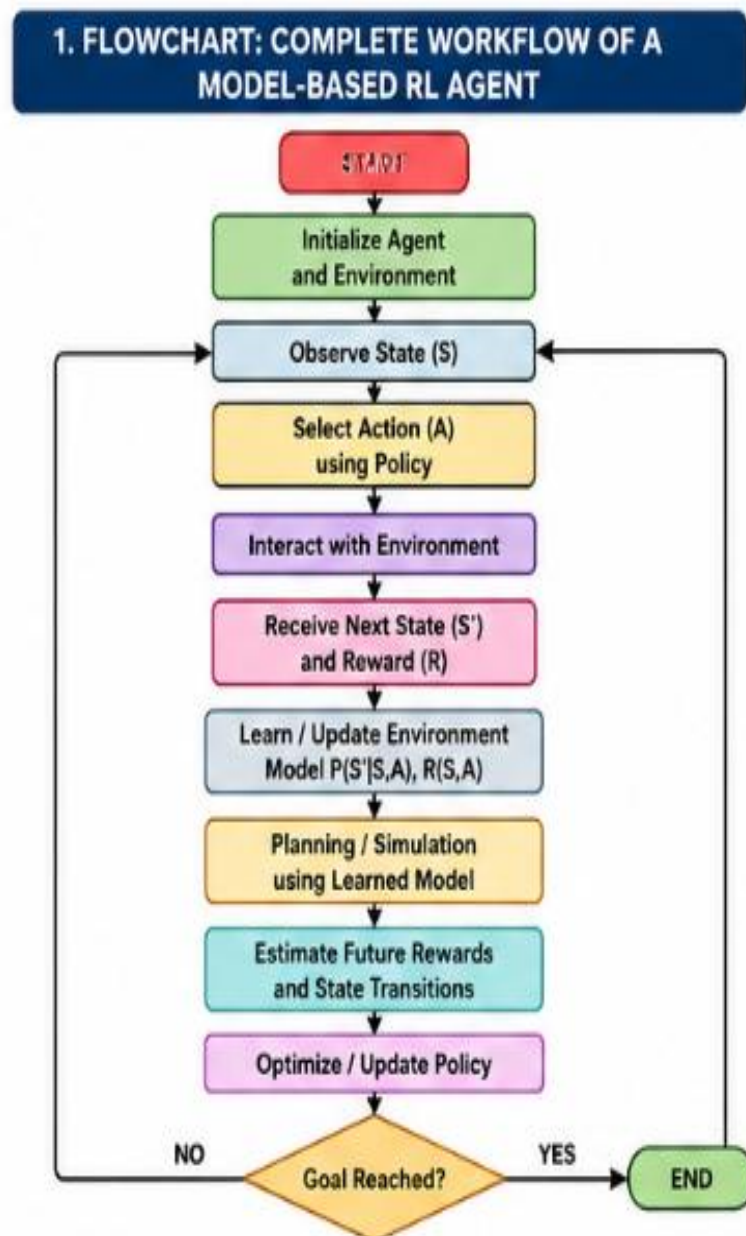
Design and Develop Model

1. Flowchart: Complete Workflow of a Model-Based RL Agent

Title: Workflow of Model-Based Reinforcement Learning Agent



FLOWCHART

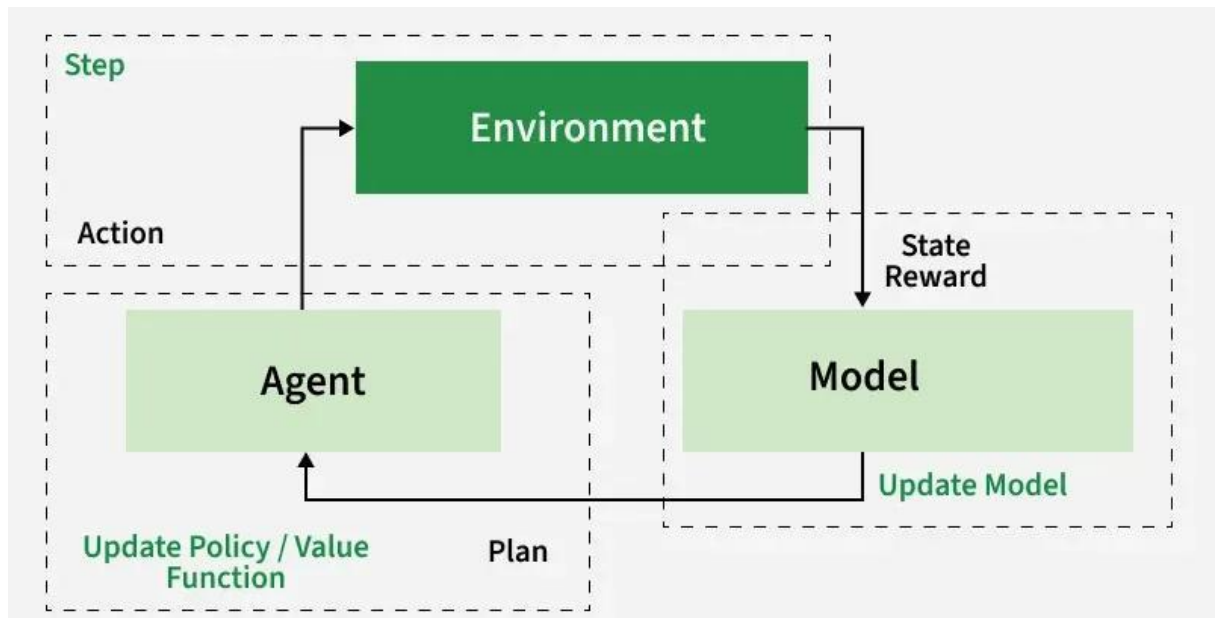


Explanation

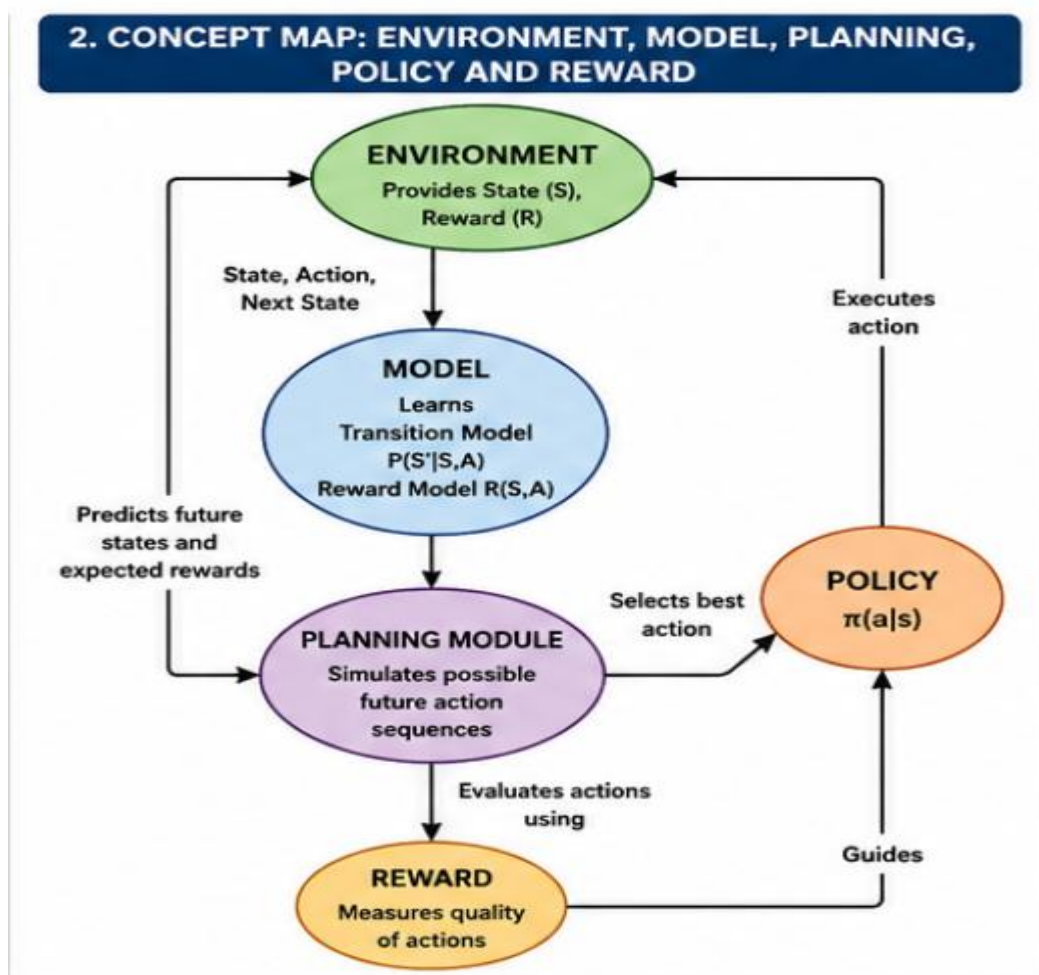
A Model-Based RL agent first interacts with the environment and collects **state**, **action**, **reward**, and **next-state information**. It uses this experience to learn an **environment model**. The learned model is then used by a **planning module** to simulate possible future actions. Based on the predicted rewards, the agent updates its policy and selects better actions.

2. Concept Map: Environment, Model, Planning, Policy and Reward

Title: Concept Map of Model-Based Reinforcement Learning



FLOWCHART



Key Relationships

Environment → provides experience → Model

Model → predicts future outcomes → Planning

Planning → evaluates actions using → Reward

Reward → guides → Policy Optimization

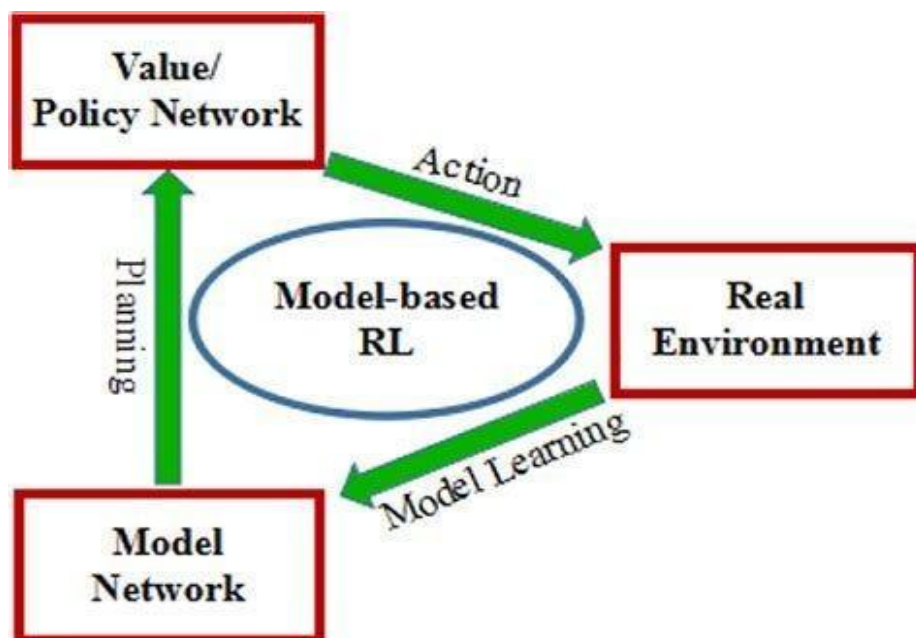
Policy → selects actions → Environment

Explanation

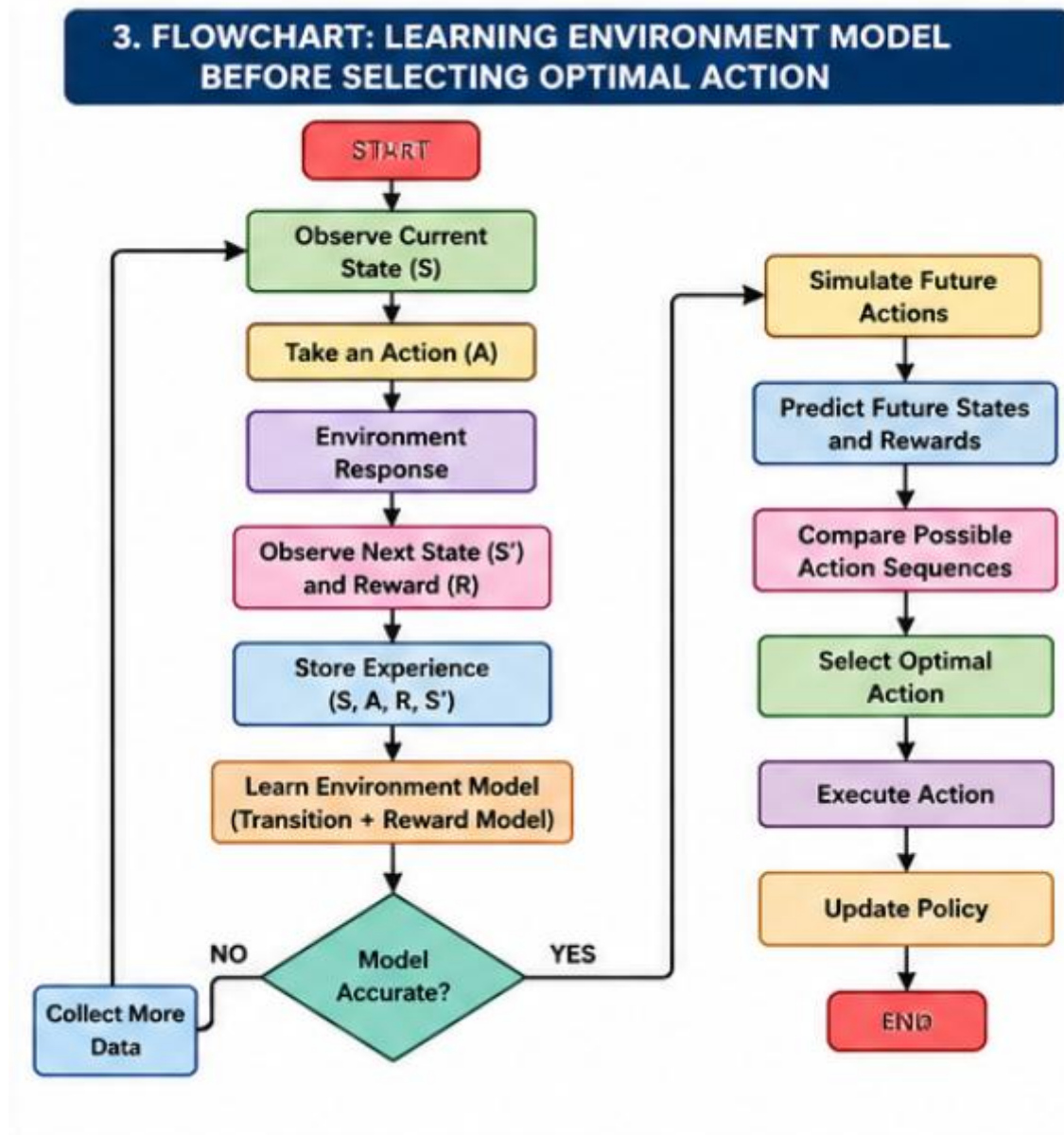
The **Environment** generates states and rewards. The **Model** learns how the environment changes after an action. The **Planning Module** uses this model to predict future outcomes.

The **Reward** evaluates the quality of actions, and the **Policy** uses these evaluations to choose better actions.

3. Flowchart: Learning the Environment Model Before Selecting the Optimal Action Title: Environment Model Learning and Optimal Action Selection



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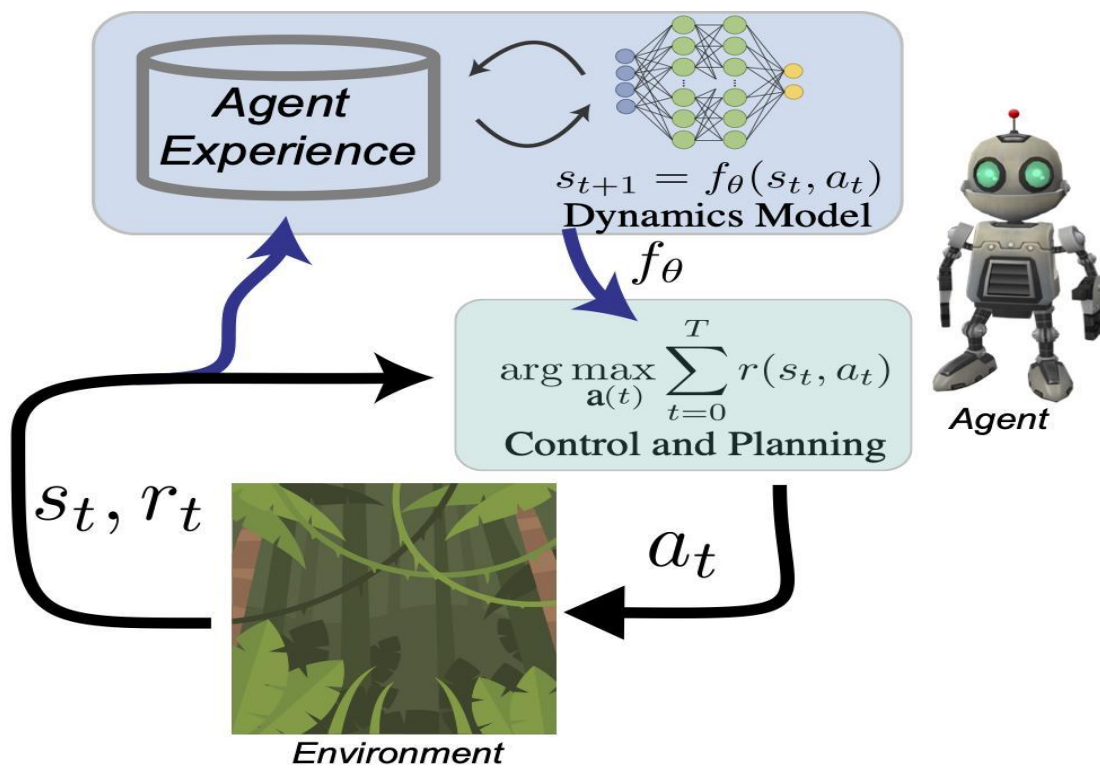


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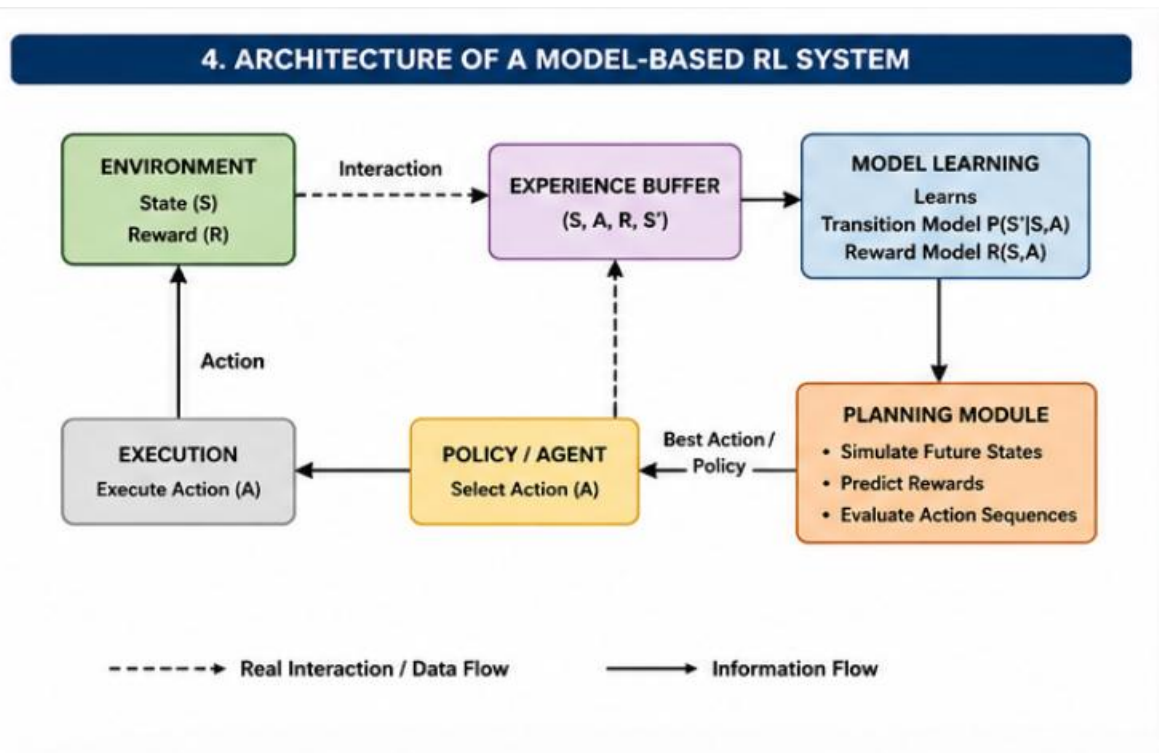
The agent first collects real experience from the environment. The experience is stored as (State, Action, Reward, Next State) and is used to learn the environment model. Once the model becomes sufficiently accurate, the agent performs simulations to predict future states and rewards. It then selects the action with the highest expected return.

4. Architecture of a Model-Based RL System

Title: Architecture of Model-Based Reinforcement Learning



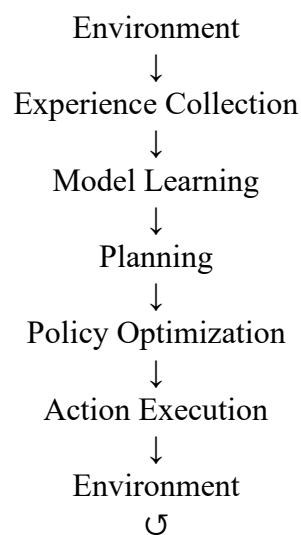
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Architecture Components

Component	Function
Environment	Provides states and rewards
Experience Buffer	Stores interaction data
Model Learning	Learns state transitions and rewards
Planning Module	Simulates future possibilities
Policy	Selects the best action
Execution	Performs the selected action
Feedback	New experience is used to improve the model

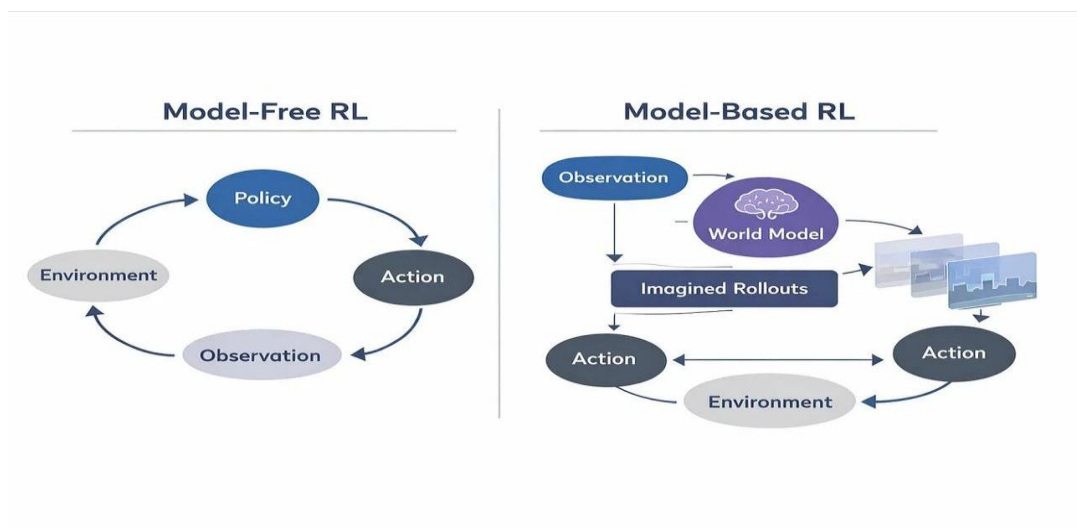
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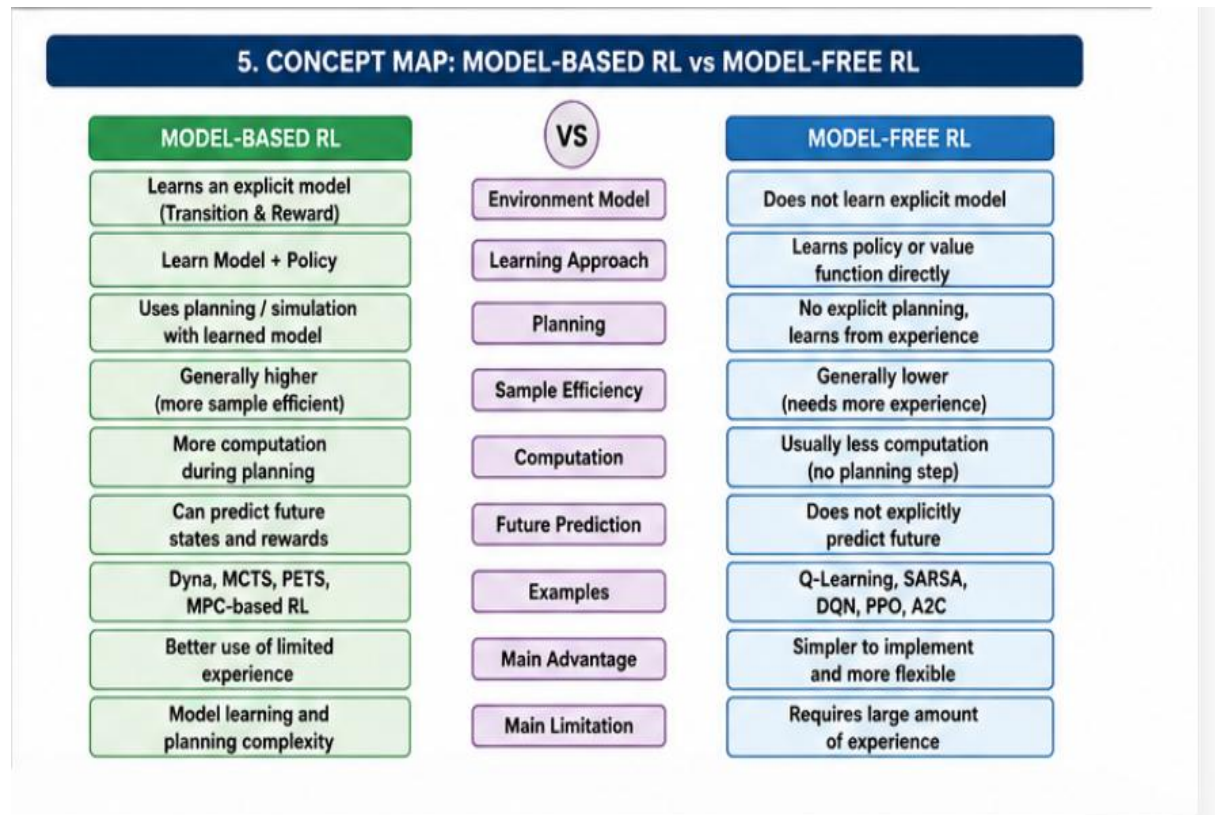
This creates a continuous learning loop.

5. Concept Map: Model-Based RL vs Model-Free RL

Title: Comparison of Model-Based and Model-Free Reinforcement Learning



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Comparison Table

Feature	Model-Based RL	Model-Free RL
Environment Model	Learns an explicit model	Does not explicitly learn a model
Planning	Uses planning/simulation	Usually learns directly from experience
Sample Efficiency	Generally higher	Generally lower
Computation	More computation during planning	Usually less planning computation
Learning	Model + policy/value	Policy/value directly
Future Prediction	Can predict future states	Does not explicitly predict them
Adaptability	Can adapt using updated model	Requires more interaction to adapt
Examples	Dyna, MPC-based RL	Q-Learning, SARSA, DQN
Main Advantage	Better use of limited experience	Simpler and often easier to implement
Main Limitation	Model-learning and planning complexity	Can require large amounts of experience

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness